

PAPER_862: Universal Magnetism U_m Fourth Master UQFF Equation

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-04 **Session:** 199 **Source:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 – Aug 07, 2025) **Calculator:** UniversalMagnetismUmMasterEquationCalc (CP4 #446) **CVW:** v2.0.0 compliant

Abstract

We formalize Universal Magnetism U_m as the fourth master equation in the quadri-adic UQFF system (alongside Compressed Gravity, Resonance, and Buoyancy). The equation $U_m = \sum_j [\mu_j(t, \rho_{vac}, [SCm]) / (r_j/r) * (1 - \exp(-\gamma t) \cos(\pi t/n)) \phi^j] * P_{SCm} * E_{react}(t) * (1 + 1e13 f_{Heaviside}) (1 + f_{quasi})$ governs magnetic and electric field dynamics through vacuum density coupling. The $\mu_j(t)$ dipole moment includes cosmic oscillation: $\mu_j = (1e3 + 0.4 \sin(\omega_c t)) * 3.38e20 \text{ T} \cdot \text{pm}^3$. The $1e13 f_{Heaviside}$ term provides the Heaviside electromagnetic amplification, while f_{quasi} captures quasi-particle corrections.

1. Core Equations

- $U_m = \sum_j [\mu_j(t)/r_j * (1 - \exp(-\gamma t) \cos(\pi t/n)) * \phi^j] * P_{SCm} * E_{react}(t) * (1 + 1e13 f_{Heaviside}) * (1 + f_{quasi})$
 - $\mu_j(t) = (1e3 + 0.4 \sin(\omega_c t)) * 3.38e20 \text{ T} \cdot \text{pm}^3$
 - $\omega_c = 1.585e-8 \text{ rad/s}$
 - $E_{react}(t) = 1e46 * \exp(-\kappa * t)$, $\kappa = 0.0005 \text{ day}^{-1}$
 - $\gamma = 0.00005 \text{ day}^{-1}$, $f_{Heaviside} = 0.01$, $f_{quasi} = 0.01$
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2. UQFF Integration

This calculator operates as a stateless physics calculator within the Condensed-Physics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py -> bodies_*.csv data flow.

3. Source Data

- **File:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 – Aug 07, 2025)
 - **Session:** 199
 - **VDS/DVP/BH:** ABSENT (general vacuum density references only)
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References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Srivastava, Y.N., Widom, A., Larsen, L. – Electroweak neutron production (LENR)
3. Kepler Mission DR25 – 4,034 candidates, 2,335 confirmed planets
4. Hubble Heritage Team / A. Nota (ESA/STScI) – Westerlund 2 / NGC 346 imaging

5. UQFF Calibration: $\kappa=0.0005/\text{day}$, $[\text{SSq}]=0.57$, $\beta_i \sim 0.603$